Beam strucutre: Wingbox

Aerospace Structures

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CONTENT

1 Introduction **2**

2 Cross-section analysis **5**

3 Beam analysis **9**

3.1 Open Section 9

3.2 Closed section 13

4 Study of the most critical point **16**

4.1 Open Section 16

4.2 Closed section 16

1 Introduction

In this third assignment a wingbox of a wing modelled as a 1D beam is studied. To do that, a cross-section and a beam analysis are carried out, for closed and open section cases.

Starting with the wingbox cross section, only the wing's central cell is taken into account and it is modelled as can be seen in figure 1. It consists of two vertical spar webs and the wing's skin itself. Both material and geometric parameters can be found on the table 1.

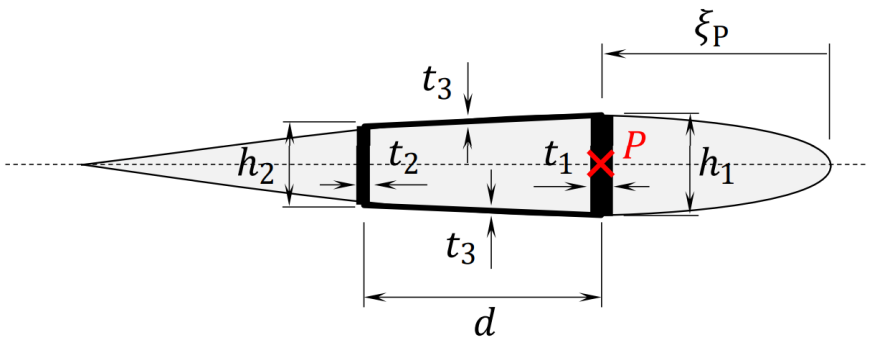

FIGURE 1: Geometry of the beam's section.

Material parameters	
E	210GPa
G	80GPa

Geometric parameters	
ξ_P	$0.3c$
d	$0.3c$
h_1	$0.25c$
h_2	$0.15c$
t_1	22mm
t_2	15mm
t_3	3.5mm

TABLE 1: Cross-section material and geometric parameters.

A cross-sectional analysis will be conducted, wherein various section characteristics such as centroid position, bending inertia, and normal stress distribution will be computed. Following that, a study on both open and closed sections will be undertaken. This study will involve calculating the torsional inertia, determining the centroid position, and analyzing the tangential stress distribution resulting from a shear load for the open section. For the closed section, the analysis will include determining the torsional inertia and examining the tangential stress distribution caused by both shear and torsion loads.

Once the cross-sectional analysis is completed, a beam analysis, and consequently a full wing analysis, will be carried on. The wing experiences a lift distribution described by expression (1). Additionally, the wing bears its own weight, uniformly distributed along its length, and at a specific point on the wing, an engine imposes a point load. All of this can be seen on figure 2, where numerical values are shown in table 2.

$$l(x) = \frac{1}{2}\rho_\infty v_\infty^2 c C_l \sqrt{1-\left(\frac{x}{b}\right)^2} \qquad \text{where} \qquad \begin{cases} \rho_\infty = 1.225\,\mathrm{kg\,m^{-3}} \\ v_\infty = 750\,\mathrm{km\,h^{-1}} \\ C_l = 0.1 \end{cases} \quad \begin{cases} c = 2\mathrm{m} \\ b = 16\mathrm{m} \end{cases} \tag{1}$$

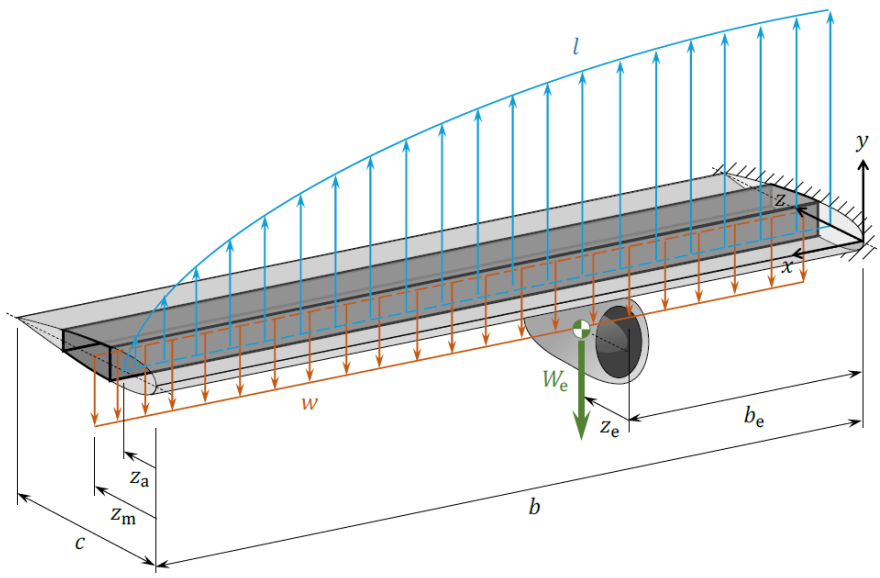

FIGURE 2: Geometry of the wing.

General parameters	
λ	$140\,\mathrm{kg\,m^{-1}}$
M_e	2100kg

Geometric parameters	
z_a	$0.25c$
z_m	$0.48c$
b_e	$0.25c$
z_e	$0.3c$

TABLE 2: Wing general and geometric parameters.

A beam analysis for both closed and open and closed section will be done, computing the different magnitudes distributions as vertical deflection, section bending rotation, section torsion rotation, shear force, bending moment and torsion moment. This will be followed with a convergence study, where the best numerical solution will be approach

using different element sizes in the beam's discretization. Finally, applying the Von Mises criterion the most critical point in the wing will be computed.

2 Cross-section analysis

In this first section a wing's cross-section analysis is carried out. The cross-sections is the one shown in figure 1, following the material and geometric parameters of table 1.

First of all, local axis must be set at some point from where different characteristic positions will be described. In this case, the lower left corner of the central wingbox cell is selected. Following that, a section discretization is done, distributing uniformly the nodes along it separated by a distance of $\Delta s = 0.00005\text{m}$ so as to get a good precision when it comes to compute the cross-section properties.

From the previously set axis, the centroid can be computed, resulting in the following coordinates:

$$x_c = 0.3987\text{m} \qquad y_c = 0.15\text{m}$$

As evident and expected, the centroid is located along the horizontal symmetrical axis. Once the centroid is computed, the inertias about it can be calculated, resulting in:

$$I_{xx} = 4.3678 \cdot 10^{-4}\text{m}^4 \qquad I_{yy} = 0.0013 \cdot 10^{-4}\text{m}^4 \qquad I_{xy} = 0\text{m}^4$$

$$J = 5.9029 \cdot 10^{-4}\text{m}^4$$

The final characteristic point that is calculated is the shear center. Considering the section is open at point P, which can be seen in figure 1, and using the centroid and the inertias previously calculated the shear center can be computed, resulting in:

$$x_s = -0.5359\text{m} \qquad y_s = 0.15\text{m}$$

Once again, the shear center is located at the horizontal symmetrical axis. This way, once all this previous calculations are done, the normal stress distribution caused by a unit bending moment about the z-axis, which is assumed to be perpendicular to the section plane, can be computed. The resulting distribution is shown in the following figure:

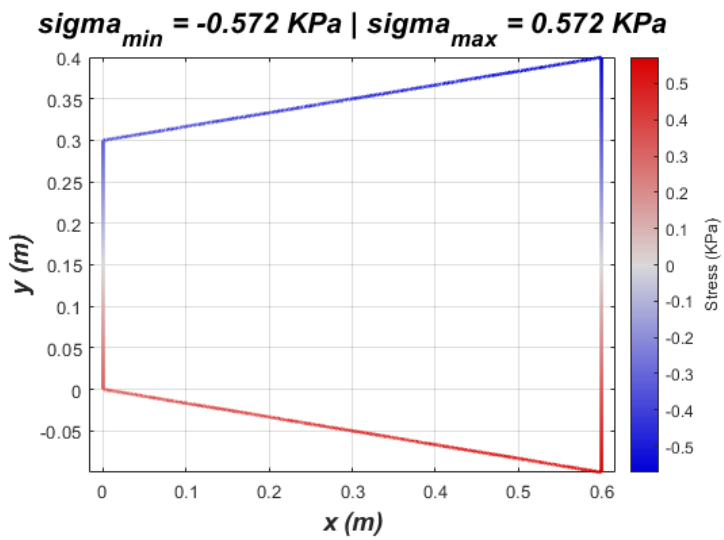

FIGURE 3: Normal stress distribution.

As can be seen and was expected, the stress distribution is totally symmetric but in opposite directions. The most solicited part are the top and bottom corners of the section, where the normal stress is 0.572kPa.

Subsequently it is assumed the section to be opened at point P. Having in mind this consideration, using the previously computed shear center and inertias, both tangential shear and torsional stress distributions, caused by a unit shear load and a torsion moment applied at shear center, can be calculated. Both distributions can be seen in the following figures:

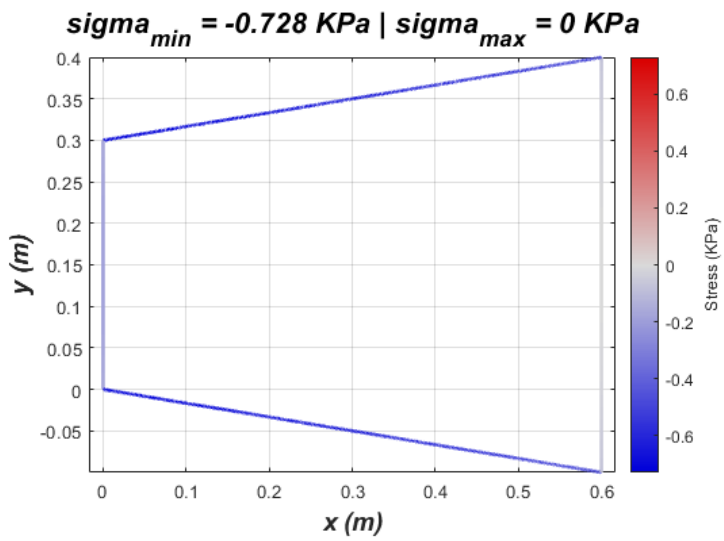

FIGURE 4: Tangential shear stress assuming the section is opened.

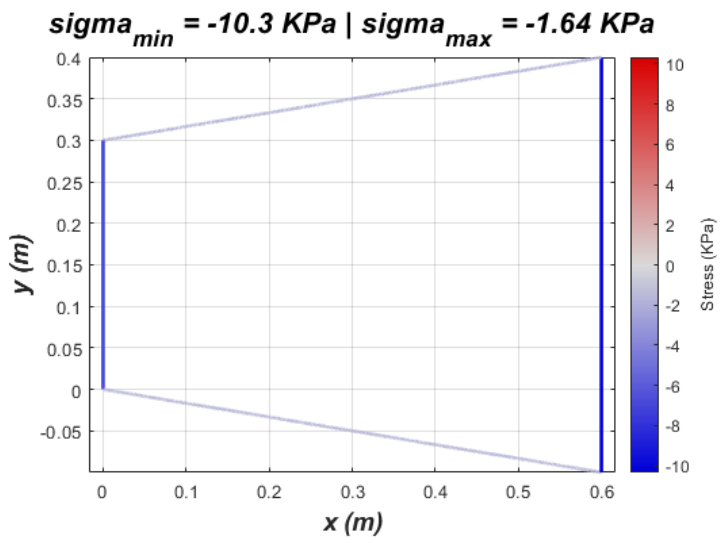

FIGURE 5: Tangential torsion stress assuming the section is opened.

As can be seen, bottom and top parts of the section are more solicited in terms of tangential shear stress, while left and right parts of the section are more solicited in terms of tangential torsion stress.

Once the open section analysis is done, the same is repeated assuming the section is closed. To achieve that, the previously calculated torsional inertia is used and the shear center is assumed to coincide with the section's centroid. That said, both tangential shear and torsional stress distributions, caused by a unit shear load and a torsion moment applied at shear center, which is the centroid, can be calculated. Both distributions are shown in the two following figures:

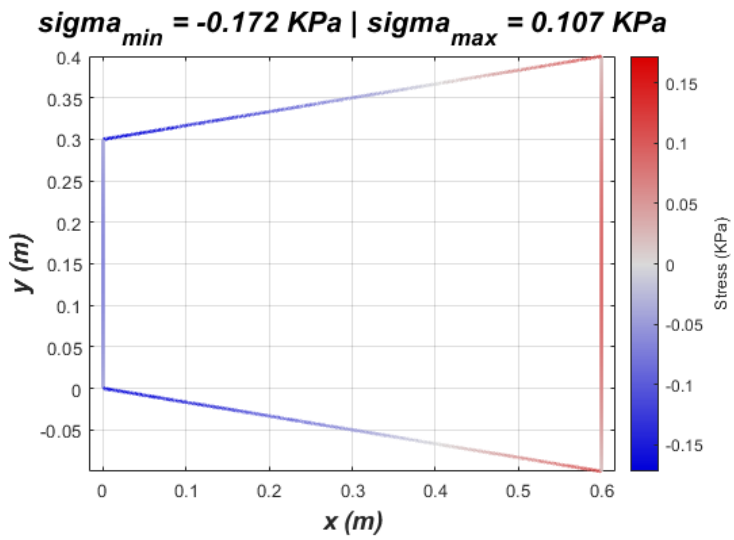

FIGURE 6: Tangential shear stress assuming the section is closed.

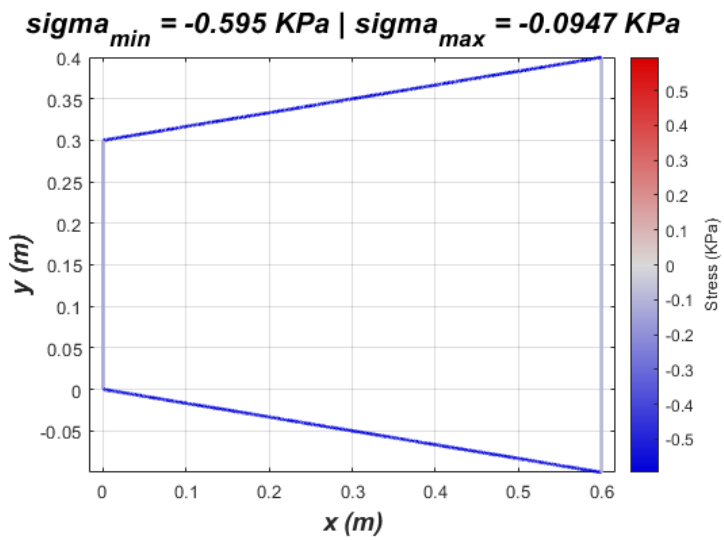

FIGURE 7: Tangential torsion stress assuming the section is closed.

As is evident, both cases, open and closed sections, exhibit significant differences. In the open section, the tangential shear stress is solely in compression, whereas in the closed section, it is present in both compression and tension, albeit with lesser stress magnitudes. Regarding tangential torsion stress, a contrasting behavior is observed between the two cases. In the open section, the stresses are greater on the left and right sides, whereas in the closed section, they are more pronounced on the top and bottom sides, albeit with significantly smaller magnitudes.

3 Beam analysis

After completing the cross-section analysis, the beam analysis should be conducted to fully understand the behavior of the wing. To achieve this, a similar procedure to the cross-section study will be followed, examining both open and closed section cases.

In the previous section, cross-section discretization was performed. In this case, a different discretization will be carried out, focusing on the wing itself. The wing will be divided into a group of elements that will assemble together to form the complete wing structure. As mentioned in the introduction, the wing experiences three types of forces: distributed lift, weight force, and a concentrated load induced by the airplane engine at a specific point on the wing. This is illustrated in figure 2. Additionally, the axis orientation will differ from that of the section analysis. In this case, the x axis will represent the longitudinal wing axis, while the z axis will define the transverse wing axis.

Now that all the previous required information has been mentioned, it is possible to numerically compute a group of magnitudes' distributions along the wing beam to study the corresponding behaviour to the different applied loads. However, it is important to note that each of the three loads is applied differently to the wing. The engine load is applied at a distance z_e from the x axis, the lift distribution is applied at a distance z_a, and the weight is applied at a distance z_m. These variations in load application are crucial for calculating different magnitudes accurately, since they'll be at the origin of a torsion moment that won't be negligible at all.

All of that said and taking into account that all of the following plots are done with a discretization of the full beam into **1024 elements**, first of all the open section case can be studied:

3.1 Open Section

In order to compute all the magnitudes of interest, it is essential to consider that for an open section, the shear center is not located at or near the centroid. This is precisely why it has been calculated previously. Following the *Appendix D. Beam Problems* document, the written code enables first of all the calculation of all three types of displacements along the wing: the vertical deflection, the section bending angle and the section twist angle.

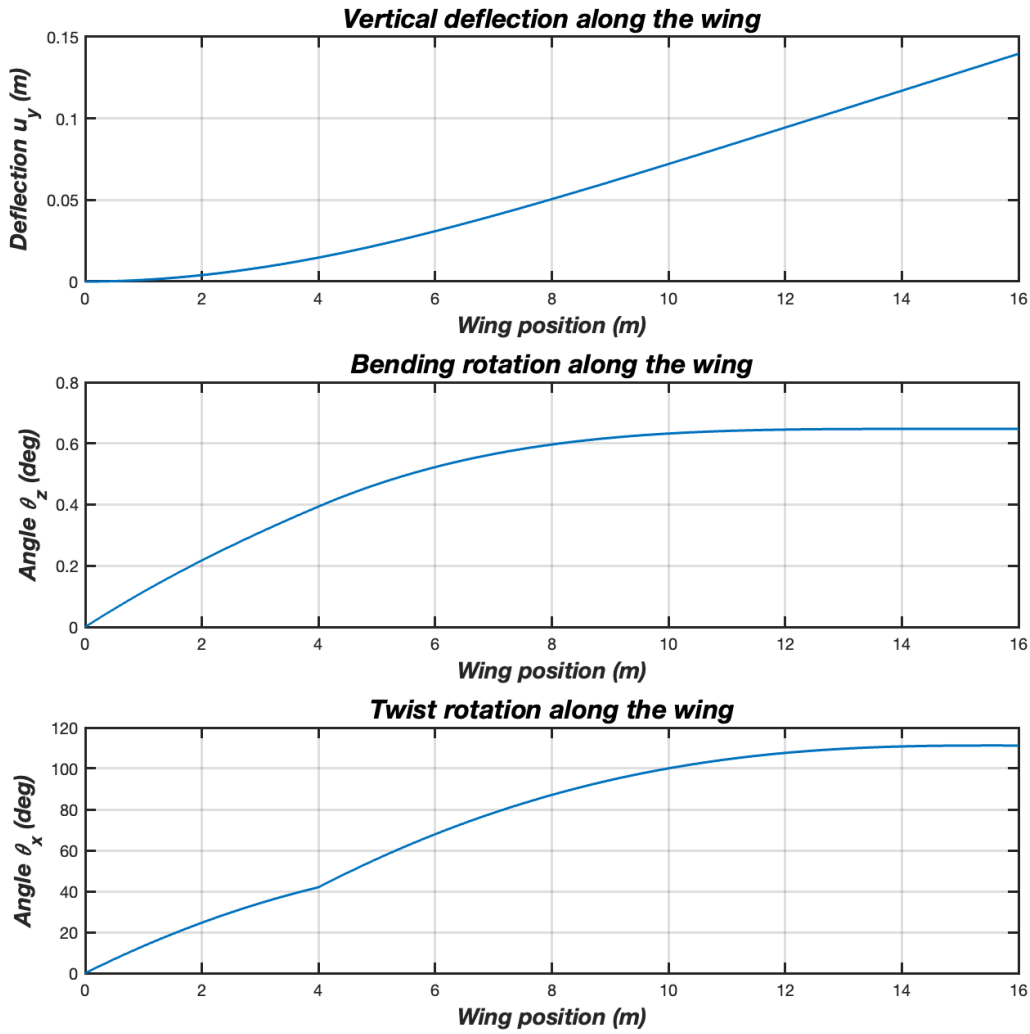

FIGURE 8: Displacements along the wing computed with 1024 panels, on open section.

Starting with the vertical deflection, a positive deflection is observed. The tip of the wing experiences the highest deflection, while the wing root remains undeflected, as expected due to the boundary condition applied. Similar behavior is observed for both bending and twist rotations. Regarding the bending rotation, it can be observed that the angle magnitude becomes constant at the end of the wing. Finally, the twist rotation plot is the only one where the effect of the engine weight is noticeable. This effect was expected, as the considerable weight of the engine and its displacement point would naturally influence the twist rotation.

Once all displacements have been plotted, the *matlab* program also enables to compute the internal forces each element of the beam is subjected to, resulting in the following three plots:

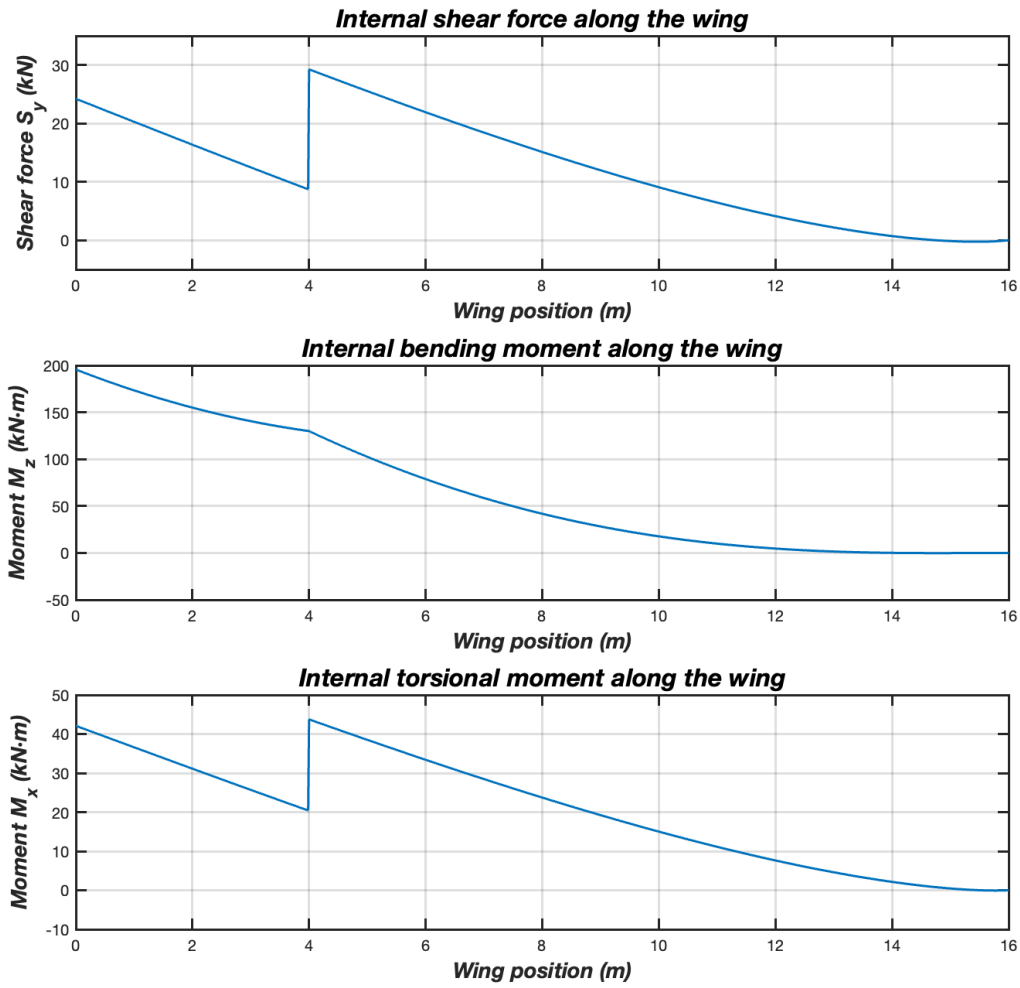

FIGURE 9: Internal forces along the wing with 1024 panels, on open section.

Regarding the internal shear force, a decreasing trend along the wing is evident, with a significant jump occurring at the engine position. This decrease in shear force can be attributed to the decreasing lift along the wing. As for the internal bending moment, a similar decrease is observed, indicating a disruption at the engine position. Finally, the internal torsional moment exhibits a similar behavior to the internal shear force, with an abrupt jump occurring at the engine position. On the three plots, at the wing tip the magnitude is zero, as it is expected.

After plotting and analyzing all these magnitude distributions, it is worthwhile to conduct a convergence study. As mentioned earlier, the previous plots were generated using a discretization of the full wing beam comprising 1024 panels. However, it is interesting to study how the computed answer changes as the number of panels increase or decreases. The convergence study is essential to know if the results are stable and reliable across different levels of discretization, allowing us to assess the accuracy and consis-

tency of the numerical calculations without the need for unnecessary computations.

In this case, the beam's discretization will utilize different element sizes h_{el}: $b/4$, $b/8$, $b/16$, $b/32$, $b/64$, $b/128$, $b/256$ and $b/512$. Subsequently, the relative errors of the deflection and twist at the wing tip will be computed for each case (open and closed section) compared to a reference discretization with a step of $b/1024$. The following expression will be used:

$$\epsilon(h_{el})\% = \left| \frac{u_{h_{el}} - u_{h_{ref}}}{u_{h_{ref}}} \right| \cdot 100 \tag{2}$$

In this way, using the previous expression the error can be expressed in a logarithmic axis, resulting in the following plot:

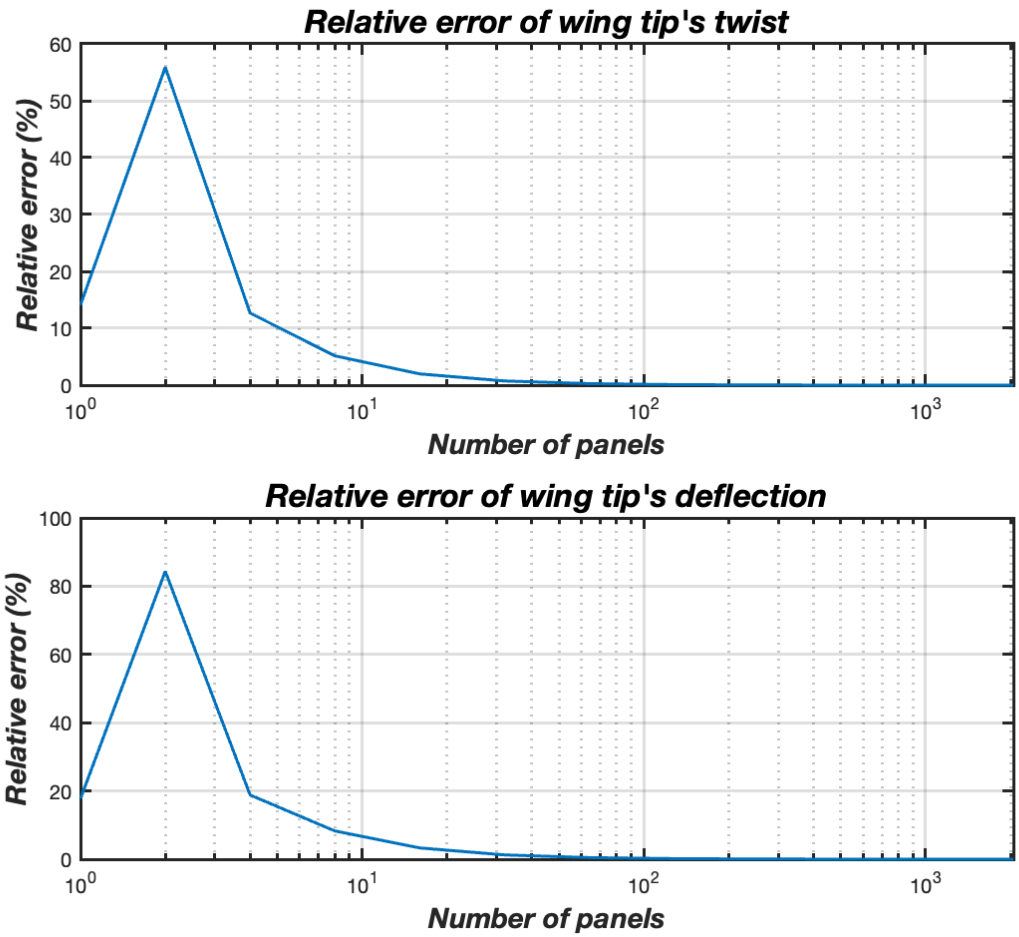

FIGURE 10: Relative error of wing tip's twist and deflection, with reference dicretization of $b/1024$, on open section.

In this last figure, it can be observed that both the wing tip's twist and deflection appear to converge at around 10^2 panels. With this number of panels, equivalent to more or less

$b/128$, it can be concluded that the computation of the different magnitudes would be sufficiently accurate, and no unnecessary calculations need to be performed.

3.2 Closed section

Once the open section study has been done, the closed section analysis must be carried out. The followed procedure is exactly the same, so only the differences between the open and closed section cases will be commented and examinated. That said, all three types of displacements along the wing are plotted resulting in the following charts:

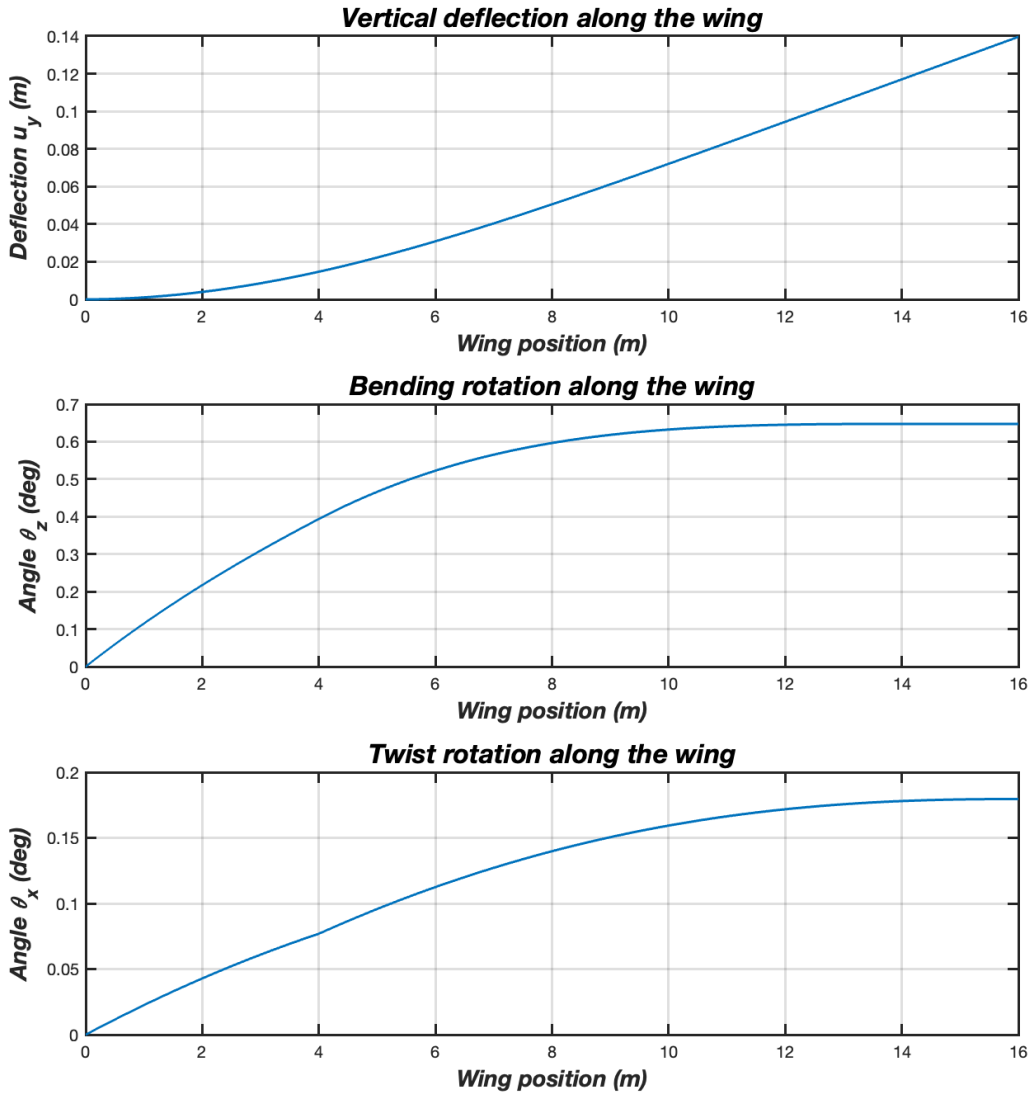

FIGURE 11: Displacements along the wing computed with 1024 panels, on closed section.

The behavior of vertical deflection and bending rotation along the wing is practically the

same as that resulting from the open sections. What truly differs is the twist rotation, which is significantly smaller in the closed section compared to the open section. This behavior was expected as the shear center is located further away from the applied forces, resulting in bigger moments.

Once the displacements have been analyzed, the same is done with the internal forces, resulting in the following plots:

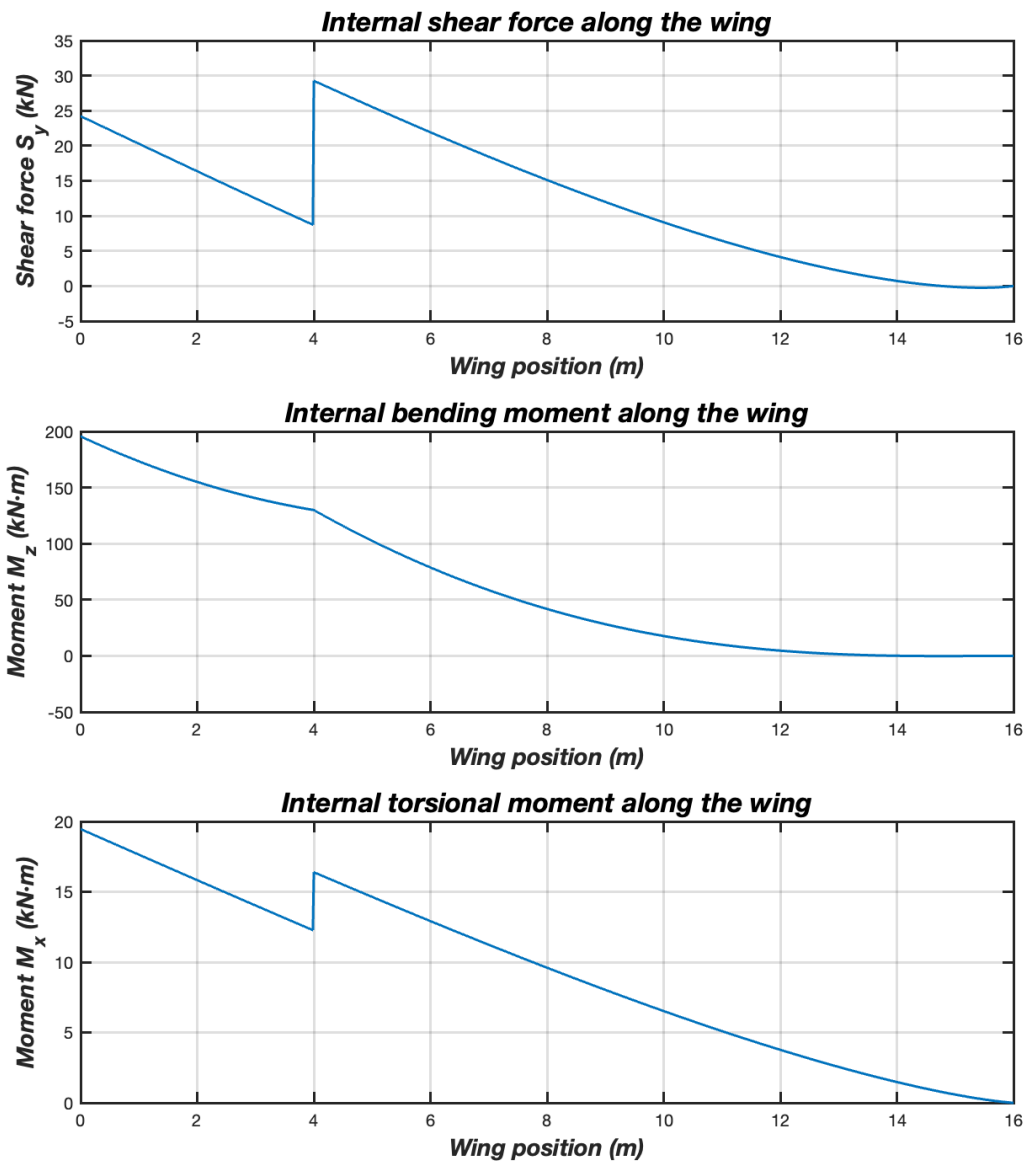

FIGURE 12: Internal forces along the wing with 1024 panels, on closed section.

With the internal forces, a similar trend is observed as with the displacements. Both internal shear force and internal bending moment are practically equal in both cases.

However, the internal torsional moment is the one that significantly differs, with considerably smaller values in the closed section case.

Finally, a convergence study is also done. Using the same expression used previously (2), the error in reference of $b/1024$ can be plotted as:

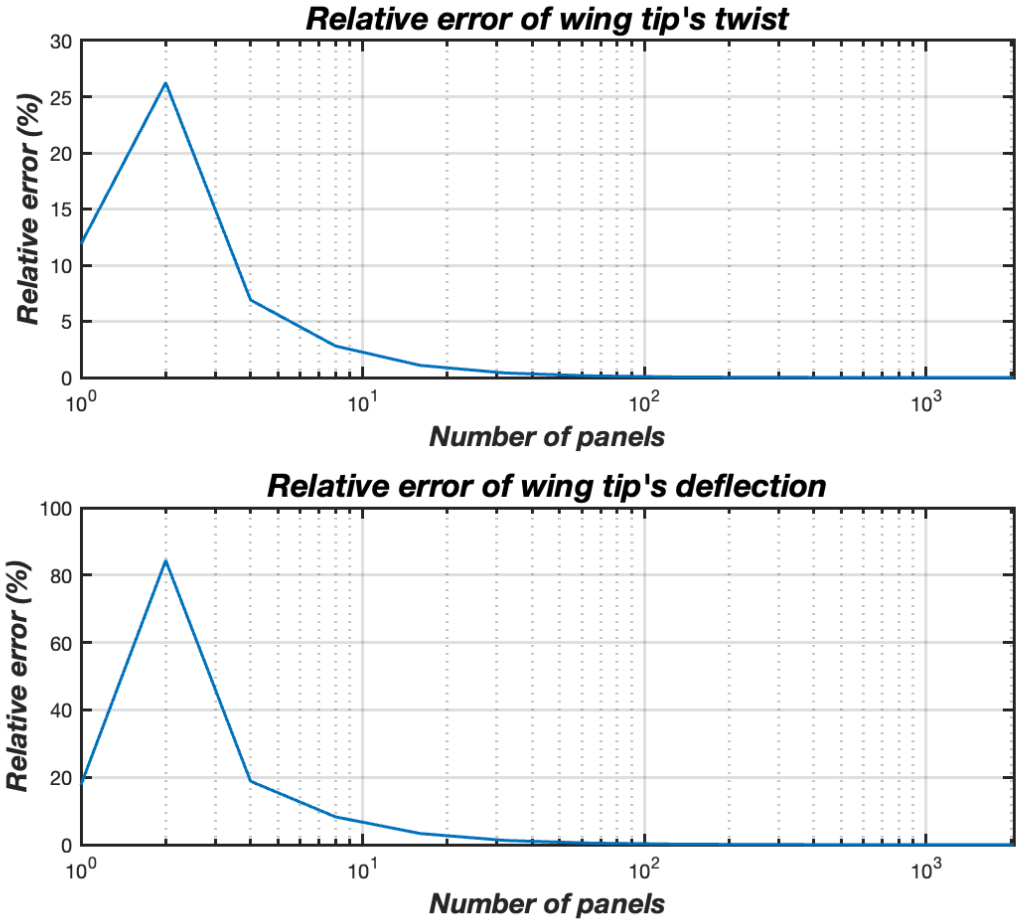

FIGURE 13: Relative error of wing tip's twist and deflection, with reference dicretization of $b/1024$, on closed section.

The same behavior observed in the open section is also observed in the closed section. With a number of panels equal to 10^2, which would represent approximately $b/128$, the error converges to zero. Therefore, starting from a number of panels equal to $b/128$, the resulting magnitudes could be considered accurate.

4 Study of the most critical point

Finally, it is interesting to identify the wing point that bears the most stress. This point may not necessarily be the same for an open section compared to a closed one, as observed, the internal forces differ. For each case, the stress on each point is computed for every section using the Von Mises (see equation (3)) criterion to calculate the equivalent stresses. Then, the maximum stress within each section is identified. After traversing all sections, the maximum of these maximum values, along with its corresponding index, is determined.

$$\sigma_{VM} = \sqrt{\sigma^2 + 3\tau^2} \tag{3}$$

4.1 Open Section

When executing the algorithm for the case of an open section, it is found that the maximum equivalent stress resides precisely at the node where the motor is applied, with a value of $\sigma_{\mathbf{VM_{max}}} = \mathbf{789.024MPa}$. Furthermore, it is located at the bottom-left corner of the section. The following figure provides a more precise visualization of the distribution of Von Mises equivalent stresses in this same section:

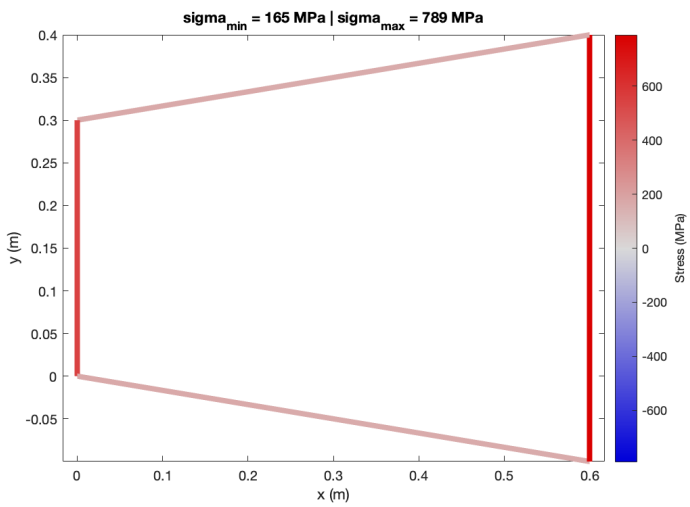

FIGURE 14: Von Mises stresses distribution at the critical section ($x = b_e$).

4.2 Closed section

On the other hand, executing the algorithm for the closed section case it can be found that the maximum equivalent stress results to appear at the wing root node, with a value equal to $\sigma_{\mathbf{VM_{max}}} = \mathbf{118.4997MPa}$. Moreover, it is located at the upper-right corner of the section. The section can be more precisely seen in the following chart, where the Von

Mises equivalent stresses are plotted.

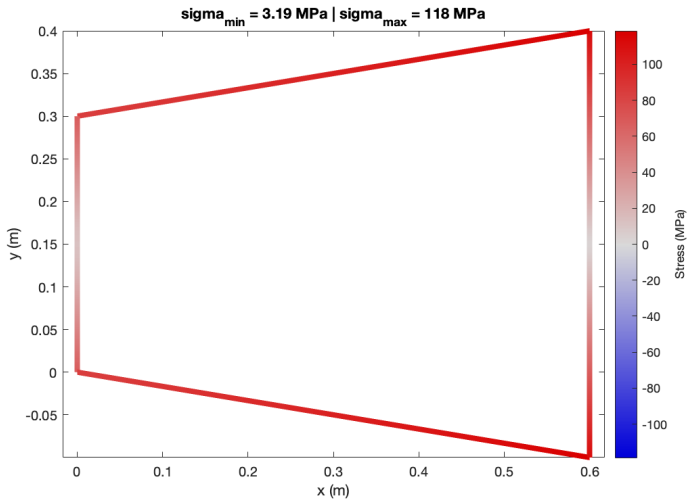

FIGURE 15: Von Mises stresses distribution at the critical section ($x = 0$).